

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

Anonymous authors
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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has increasingly become a consequential paradigm for strengthening the reasoning competence of large language models, particularly because it permits optimization from externally checkable outcome signals rather than from costly preference annotations. Nevertheless, extant approaches such as GRPO frequently encounter zero-gradient regimes, a pathology that arises chiefly from fixed clipping constraints on token-level probability ratios and from the stepwise standardization of identical rewards, both of which can attenuate effective gradient transmission and leave a substantial fraction of generated responses unused. In this work, we propose **Dynamic Clipping Policy Optimization (DCPO)**, which combines a dynamic clipping strategy that adjusts admissible ratio bounds according to token-specific prior probabilities, thereby enlarging token-level exploration, with a smooth advantage standardization procedure that normalizes rewards across cumulative training steps, thereby improving response-level utilization of generated data. Across four benchmarks and four model backbones, DCPO achieves state-of-the-art performance. In particular, on Qwen2.5-Math-7B, DCPO obtains an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-fold sampling on AIME24, exceeding DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On AIME25 with Qwen2.5-14B, DCPO reaches (23.3/19.0), outperforming GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). Moreover, DCPO improves the nonzero-advantage proportion by an average of 28% over GRPO across four models, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude compared with both GRPO and DAPO while maintaining superior task performance. Collectively, these findings indicate that DCPO leverages generated trajectories more efficiently for reinforcement learning in large language models.

1 INTRODUCTION

In recent years, the reasoning capabilities of large language models (LLMs) have attracted sustained and intensifying attention (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) has correspondingly emerged as a particularly promising mechanism for improving such capabilities, since it optimizes LLMs with rule-based outcome rewards and is therefore well suited to domains, such as mathematics and coding, in which reflective decision-making can be externally verified (DeepSeek-AI et al., 2024).

To operationalize RLVR, the GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms has been developed and widely adopted. Although the empirical robustness of GRPO has been corroborated across numerous studies, subsequent work, including DAPO (Yu et al., 2025), has exposed important limitations: GRPO can undergo entropy collapse, manifested as the decay of advantage values to zero in later training stages, and inappropriate weight clipping can suppress rarely sampled yet informative learning signals, thereby unduly narrowing the diversity of policy exploration.

DAPO attempts to alleviate these difficulties by introducing Clip-Higher, which relaxes the upper clipping bound to counteract entropy collapse, and Dynamic Sampling, which filters samples with uniform rewards so that each batch contains only responses associated with valid gradients. However, these remedies retain two salient deficiencies: (i) **Response-Level Inefficiency**: dynamic sampling necessarily lowers sampling efficiency and consequently slows convergence; and (ii) **token clipping**

deficit: our analysis indicates that the appropriate clipping bound is token-dependent, rendering any uniform upper bound intrinsically suboptimal. Although alternative designs may avoid the direct removal of zero-gradient responses, these limitations remain structurally persistent.

In this work, we introduce **Dynamic Clipping Policy Optimization (DCPO)**, a policy-optimization framework explicitly designed to address the central deficiencies of prior methods. First, DCPO incorporates a **smoothing technique that standardizes advantages across cumulative steps**, thereby mitigating the gradient-loss problem induced when responses receiving identical rewards are assigned zero advantages. By integrating the reward distribution of all cumulatively generated responses with the distribution produced at the current step, this procedure enables more efficient response-level exploration (see Section 2). Second, rather than specifying distinct but still fixed upper and lower bounds, DCPO employs a **dynamic clipping method that adaptively adjusts the clipping bounds based on the old token-specific probabilities**, thereby addressing the rigidity of fixed clipping. Concretely, lower-prior-probability tokens receive wider admissible clipping intervals, which expands the space for token-level exploration (see Section 2 and Appendix A.6). In addition, DCPO computes the mean loss only across tokens within each individual response, rather than over the entire batch, so that the relative advantage structure among responses to the same prompt is preserved and the dilution induced by batch-level averaging is avoided.

For evaluation, we merge Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of the MATH dataset (Liu et al., 2025) to construct the training corpus, and we assess performance on four standard mathematical reasoning benchmarks. Experiments across four model backbones compare DCPO against GRPO, DAPO, and, where available, GSPO. The results show that DCPO consistently yields superior preference-optimization outcomes, with especially pronounced gains on the challenging AIME benchmarks: under 32-time sampling on AIME24-Avg@32, DCPO-7B scores 38.8, compared with 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO), whereas on AIME25-Avg@32, DCPO-14B reaches 19.0, compared with 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). Moreover, DCPO raises the average nonzero-advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude compared with all baselines while delivering stronger task performance.

Additional preliminary material, including the loss formulations and advantage-calculation procedures used by prior algorithms such as GRPO, DAPO, and GSPO, is provided in Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Probability-ratio clipping is a canonical stabilization device in reinforcement learning, where it limits excessively large parameter displacements during policy optimization. Its importance is especially pronounced in off-policy regimes that rely on importance sampling: reweighting by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$ yields an unbiased estimate of $\mathbb{E}_{q(x)}[f(x) \frac{p(x)}{q(x)}]$, yet this estimator can incur substantial variance inflation, as characterized in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To attenuate this inflation, prior work (e.g., Schulman et al. (2017)) constrains the probability ratio within a fixed interval governed by the hyperparameter ϵ , as specified in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping interval, however, restricts exploration in the diversity space, especially in regions to which the old policy assigns little probability mass. In such regions, the absolute admissible change for decreasing probabilities is intrinsically narrow, which curtails meaningful policy updates. Although methods such as DAPO introduce asymmetric clipping bounds to partially address this issue, the underlying non-adaptivity remains; moreover, GSPO discards entire responses once the sequence-level importance ratio exceeds the asymmetric bounds, thereby wasting informative tokens while retaining additional high-variance tokens that can destabilize training.

From the standpoint of variance–bias control, a more operationally appropriate strategy is to impose probability-dependent constraints, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3), and enforcing non-negativity of the square-root term by taking its maximum with zero, gives a closed-form solution for $r(x)$. The resulting **dynamic-adaptive clipping bounds** vary as functions of $q(x)$ and of the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive clipping mechanism therefore permits substantially greater policy exploration in low-probability regions. Furthermore, following the intuition of dual clipping (Ye et al., 2020), we cap the maximum bounds for both positive and negative advantages at 10, preventing excessively permissive clipping intervals from inducing overtraining. A detailed derivation and analysis of these dynamic clipping bounds within the importance-sampling framework is provided in Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior methods such as GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ assigned to the j -th response is obtained by standardizing only the rewards $R_{j=1,\dots,G}^i$ associated with the same prompt at the i -th step, as shown by Equation (11) in Appendix A.3. This purely step-local normalization can be problematic: when stochastic response sampling produces identical rewards at a given step, all advantages collapse to zero, preventing the prompt from contributing to parameter updates despite possible variation in reasoning trajectories. Moreover, high-entropy sampling can produce strongly skewed label counts, which in turn causes large cross-step fluctuations in standardized advantage values, including sign reversals that destabilize optimization.

At the same time, recent work (Yue et al., 2025) suggests that RLVR primarily refines an existing policy by adjusting the likelihood of correct samples, rather than fundamentally reconstituting the underlying distribution. Under this view, the reward distribution of responses to a given prompt can be treated as being drawn from a shared global distribution throughout training, which motivates the *cumulative* standardization in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the corresponding statistics are computed over all responses generated so far for the same prompt.

To reduce fluctuations arising from both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, which instantiate weighted averages of the two normalization schemes with step-dependent weights, as defined in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ increasingly emphasizes the current distribution as training proceeds, thereby privileging adaptation to the most recent policy outputs. Conversely, $\hat{S}A_{total,j}^i$ assigns greater weight to the cumulative distribution, treating the current-step deviation as a local adjustment within a broader empirical reward landscape. The final advantage $\hat{A}_j^i$ is then chosen as the smoothed advantage with the smaller absolute magnitude, which reduces the destabilizing influence of fluctuations introduced

by either cumulative or current-step standardization, as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once a prompt has entered model optimization, responses generated for that prompt in subsequent steps continue to participate in optimization regardless of whether the current-step advantage is zero. Consequently, even when identical rewards arise within a step, the corresponding responses update the model through the advantage $\frac{1}{i}\hat{A}_{total,j}^i$, which preserves otherwise lost learning signal and improves data efficiency.

3.1 OTM LOSS

Although the **Token-Level Mean loss (TLM)** (the loss Equation (13) used in DAPO) equalizes token contributions, it can distort the relative importance of responses with different lengths. Consider, for example, two responses to the same prompt: response *A* has an advantage of 1 and length 500, whereas response *B* has an advantage of 0.5 and length 1500. Under TLM, *A* receives weight $\frac{500}{500+1500} = 0.25$, while *B* receives weight $\frac{1500}{500+1500} = 0.75$; hence their total loss contributions become $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, respectively. The lower-advantage response *B* therefore exerts greater influence (0.375) solely because it is longer.

Conversely, when averaging across batches, the **Sequence-Level Mean loss (SLM)** (the loss Equation (12) used in GRPO) attenuates advantage magnitudes, effectively reducing them by a factor of *G* after advantage computation. This attenuation is especially damaging for large batches, because it erodes the standardized relational structure among the original advantages. Since standardized advantages already encode relative relationships across responses, averaging over all tokens within the same response by $|o_i|$ both preserves those relationships and allocates importance evenly across the tokens of the current response. Accordingly, we average tokens only within a single response, without additionally averaging across all responses in the same batch; this objective is termed **Only Token Mean loss (OTM)** and is given in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, the DCPO loss is not constrained by an explicit KL-divergence term, and its lower and upper clipping bounds are the dynamic clipping bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets We conduct experiments with four model backbones—Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b)—and train on an approximately 25k-problem mathematical corpus formed by merging DAPO-Math-17K (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). For both training and evaluation, every problem is instantiated with the Qwen-Math template (see Section A.8).

Evaluation We evaluate the resulting models on four widely used mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). Because **MATH500** contains 500 problems, we report only Avg@1 for that benchmark; for the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) datasets, we report both Avg@1 and Avg@32, thereby capturing performance under deterministic and sampling-based decoding.

- **Avg@1:** This metric denotes the standard accuracy obtained under greedy decoding and measures the model’s single best prediction.
- **Avg@32:** This metric reports the average accuracy over 32 sampled responses per problem, using temperature 1.0 and top-p 1.0, and therefore characterizes the robustness and stability of the trained policy distribution.

Baseline Settings We use GRPO and DAPO as primary baselines and additionally include GSPO for Qwen2.5-Math-7B and Qwen2.5-14B. Other configuration details, including method-specific clipping thresholds ϵ_* , the training codebase, and shared hyperparameters, are documented in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1: Benchmark performance under greedy decoding (Avg@1) and 32-sample decoding (Avg@32). Boldface denotes the best value in each comparison, and Δ reports improvement over the corresponding baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

Table 1 summarizes the performance of GRPO, DAPO, GSPO, and the proposed DCPO across four mathematical reasoning benchmarks: MATH500, AMC23, AIME24, and AIME25. By reporting both greedy decoding accuracy (Avg@1) and sampling-based accuracy (Avg@32), the table jointly evaluates deterministic performance, robustness, and generalization under stochastic decoding. Across all model scales, DCPO consistently delivers superior or comparable results relative to GRPO, DAPO, and, where available, GSPO, indicating that it functions effectively as a general-purpose preference-optimization method for reinforcement learning with LLMs.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO achieves the highest average scores (47.2/32.8), exceeding GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO obtains (37.6/22.7), improving over GRPO (+1.3/+1.7) and remaining competitive with

DAPO, with a clear Avg@1 gain (+1.0) and only a marginal Avg@32 difference (-0.4). For Qwen2.5-Math-7B, DCPO reaches (57.1/45.2), surpassing GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On the largest evaluated model, Qwen2.5-14B, DCPO achieves (53.2/39.0), yielding substantial gains over GRPO (+6.5/+7.7) and outperforming DAPO by (+1.3/+0.1) and GSPO by (+4.2/+5.5) in Avg@1 and Avg@32. Taken together, these results demonstrate that DCPO produces robust improvements across model capacities and aligns more effectively with the reward signal under both greedy and stochastic decoding regimes.

Robustness on AIME under Sampling (Avg@32) A particularly salient pattern appears on AIME24 and AIME25, two high-difficulty competition mathematics benchmarks designed to stress-test model reasoning. Under Avg@32, which measures the capacity to generate correct answers under high-entropy sampling, DCPO exhibits markedly stronger robustness at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B achieves 38.8, outperforming GRPO (32.1), DAPO (31.6), and GSPO (34.9), while on AIME25-Avg@32, DCPO-14B reaches 19.0, exceeding GRPO (10.5), DAPO (15.3), and GSPO (9.9). These gains indicate that DCPO improves not only average-case accuracy but also the diversity and reliability of correct generations, an especially important property for sampling-based decoding where temperature-driven variability produces multiple plausible outputs. Its simultaneous optimization of reward fidelity and exploration under high-entropy decoding therefore underscores its practical value.

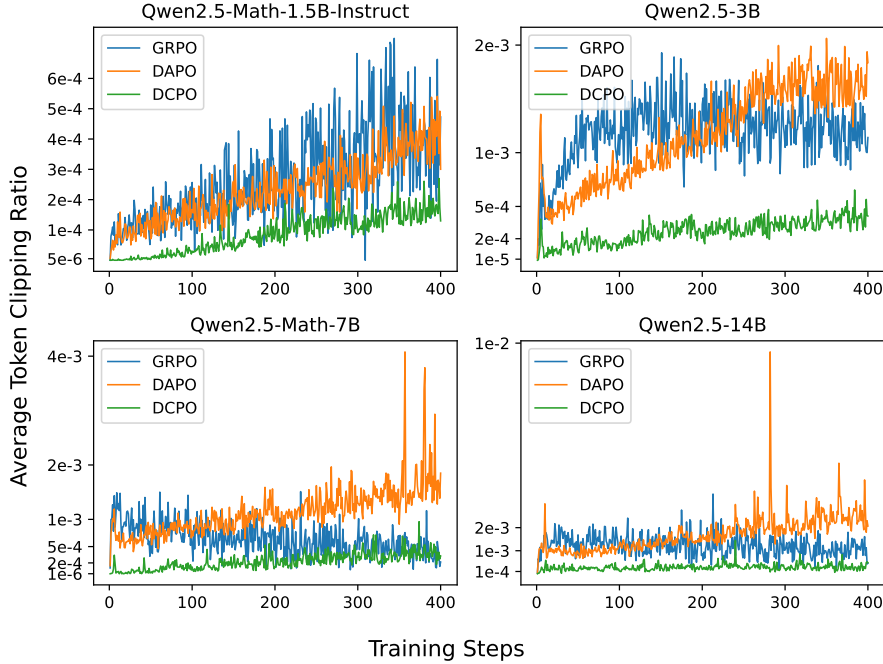


Figure 1: *TCR* trajectories across models and optimization methods.

5.2 THE TOKEN CLIPPING RATIO (*TCR*)

We define the **Token Clipping Ratio** (*TCR*) as the proportion of tokens excluded from policy updates during back-propagation because their ratios are clipped. As shown in Figure 1, *TCR* dynamics differ substantially across reinforcement-learning methods.

$$\text{TCR} = \text{Average} \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } \textit{micro}_m}{\text{Total number of tokens in } \textit{micro}_m} \right) \quad (9)$$

where $\textit{micro}_m$ denotes the m -th microbatch and N denotes the number of microbatches in each training step.

During GRPO training, *TCR* exhibits model-size-dependent behavior: for smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B), it increases across training steps, whereas for larger models (Qwen2.5-

Math-7B and Qwen2.5-14B), it gradually declines. DAPO, by contrast, displays a consistently increasing TCR across all models, implying that progressively more tokens are clipped and that a larger fraction of policy updates is computed from partial or truncated responses. Both GRPO and DAPO also show pronounced fluctuations and occasional abnormal spikes in TCR , which may inject instability into optimization and contribute to entropy collapse. Meanwhile, GSPO’s average TCR exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B, substantially above the average TCR of token-level clipping methods and therefore wasteful of informative tokens.

In contrast, DCPO maintains a substantially more stable and efficient training regime. Across model sizes and training stages, its TCR remains nearly constant and is approximately an order of magnitude lower than those of GRPO and DAPO. This lower and more stable TCR indicates that DCPO discards fewer tokens, permits a larger portion of complete response sequences to influence policy updates, and reserves additional exploratory capacity for rare-token diversity. The result highlights DCPO’s superior sample efficiency while preserving control over outlier variation.

Using Qwen2.5-Math-7B as an illustrative case, we observe rapid concentration of the token distribution on high-confidence tokens: after approximately 60 steps, about 95% of generated tokens have probability ($q(x) > 0.9$), and this fraction rises to roughly 97% after approximately 100 steps before continuing to increase later in training. As illustrated in Figure 4d, once the old probability satisfies ($q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$), the new probability $p(x)$ participates in model updates within the interval $[\frac{1}{1+\epsilon}, 1]$ without being discarded by any clipping method. Consequently, most tokens occupy a never-clipped region, explaining the low aggregate TCR observed empirically. At the same time, Figure 1 shows that TCR increases during later phases of most training runs, suggesting that a growing share of low-probability tokens may be discarded and that exploration of rare-token diversity is correspondingly constrained.

Prior work (Wang et al., 2025) demonstrates that **high-entropy tokens**, which correspond precisely to low-probability tokens, are primary drivers in the emergence of model reasoning capabilities. Because entropy and probability are inversely related, rarer tokens carry greater informational content; consequently, granting these low-probability tokens a broader admissible update range enlarges the model’s effective exploration space and facilitates the acquisition of diverse reasoning behaviors.

These empirical and theoretical observations motivate the proposed **dynamic-adaptive clipping** scheme, which relaxes clipping for tokens satisfying $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens. By adaptively widening the update interval over the high-entropy tail of the distribution, the method restores the model’s ability to learn from rare, informative tokens and thereby strengthens reasoning performance.

5.3 THE RESPONSE UTILIZATION RATIO (RUR)

During RLVR training, a response contributes to policy optimization only when it yields a nonzero gradient. We quantify this utilization efficiency with the **Response Utilization Ratio (RUR)**, defined as the percentage of generated responses with nonzero advantage that participate in policy updates.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 shows that the average RUR for responses generated during GRPO training is low, with an overall mean of only 44.6% across four models. Thus, more than half of generated responses are effectively discarded, revealing substantial inefficiency in response utilization during diversity exploration. By contrast, DCPO achieves an average RUR of approximately 70% after the first epoch, corresponding to an absolute increase of 28% and a relative improvement of 64% over GRPO.

Table 2: Average RUR across 400 training steps.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that GRPO’s *RUR* declines substantially over time, irrespective of model size or type. In particular, *RUR* falls sharply from above 90% to below 50%; for Qwen2.5-Math-7B, it drops to approximately 30% after around 200 training steps. This pattern suggests that GRPO’s reliance on current-step advantage standardization creates a large fraction of zero-gradient responses, making diversity exploration increasingly inefficient in later training. By contrast, DCPO, through smooth advantage standardization, consistently sustains a high *RUR*; after the first epoch, *RUR* rapidly stabilizes around 70% and continues to improve as training proceeds.

We observe that *RUR* exceeds 90% during the first few steps, but from the second epoch through the end of training it remains near 70% rather than approaching 100%. We analyze these phenomena to clarify the potential value of the SAS mechanism.

First, the initial *RUR* above 90% is largely attributable to early-stage outputs that violate the required format, receive negative rewards, and thereby inflate *RUR*. This behavior indicates that different models rapidly acquire fixed answer-format patterns, after which training should emphasize the exploration of substantive reasoning capabilities. Second, our analysis identifies two principal sources of persistently zero advantages, which explain why *RUR* stabilizes near 70% rather than reaching 100%: (i) nearly half of the math problems are sufficiently easy that the model solves them consistently, making their exclusion from further policy updates reasonable because the relevant capabilities require little additional refinement; (ii) the remaining cases are consistently erroneous. This persistence may arise because some problems exceed the current model’s solving capacity even when the ground-truth label is correct. Alternatively, DAPO-Math-17K contains items whose labels were converted into mislabeled integers, creating potentially “incorrect” labels. In such cases, discarding zero-advantage responses functions as a filtering mechanism that reduces the risk of updating the model on noisy or mislabeled data.

Unlike GRPO and DCPO, DAPO removes generated responses with identical labels, which makes direct *RUR* comparison difficult. We therefore evaluate DAPO’s training efficiency by comparing the number of generated responses and the training time required to complete the same number of update steps. To compensate for the low sampling efficiency induced by GRPO’s low *RUR*, we set DAPO’s generation batch size to 1–4 times the training batch size and adjust it dynamically during training. Across four models, DAPO requires 3 to 5 times more generated responses than DCPO to obtain the same number of parameter updates, at least doubling GPU hours without improving performance. Moreover, DAPO’s data-utilization efficiency declines with model size; for example, on Qwen2.5-14B, the proportion of responses used by DAPO falls below 30% after approximately 300 steps. These results indicate that DAPO’s dynamic sampling approach is inefficient in both data utilization and training cost, while also forfeiting opportunities for diversity exploration.

In summary, DCPO constitutes a substantial improvement over both GRPO and DAPO by increasing response utilization, improving training efficiency, exploiting sampled responses more completely, expanding the exploration space for model diversity, and ultimately improving performance.

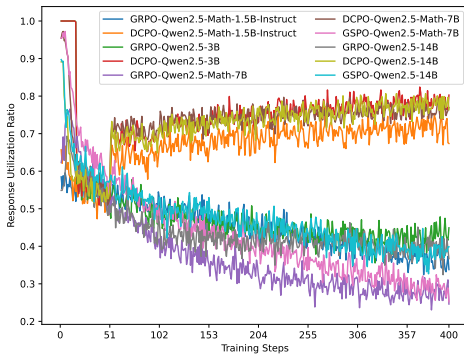


Figure 2: *RUR* evolution throughout training.

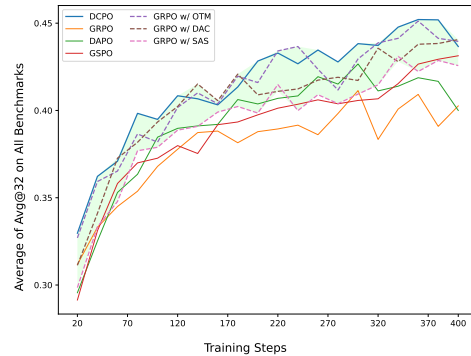


Figure 3: Ablation results measured by average Avg@32 on Qwen2.5-Math-7B.

5.4 ABLATION STUDIES

We conduct an ablation study on Qwen2.5-Math-7B to isolate the contribution of each DCPO component, using Avg@32 as the evaluation metric because it emphasizes the robustness and stability

of the learned policy distribution. To preserve comparability, each experiment modifies only one component of the baseline GRPO framework, keeps all remaining settings fixed, and removes the KL-divergence term so that the configuration aligns with DAPO, GSPO, and full DCPO. The learning curves from 20 to 400 steps are reported in Figure 3.

The learning curves for the three individual components indicate continuous improvement in inference ability: each variant surpasses GRPO and reaches, or exceeds, the performance of DAPO and GSPO. Thus, every modification contributes a distinct benefit, while their combination yields a substantial aggregate gain over the baselines, demonstrating the effectiveness of the DCPO pipeline for reinforcement-learning optimization. In later training stages, baseline performance plateaus, whereas DCPO continues to trend upward both for individual components and for the complete method. The ablation results therefore confirm that each component improves training efficiency and that their integration produces cumulative gains and enhanced stability. Detailed component-level observations are as follows.

(i) **Only-Token-Mean loss (OTM)**: When GRPO aggregates loss within each individual response rather than across the entire batch, the resulting model outperforms baselines that exhibit weaker data utilization and more volatile performance. (ii) **Smoothed Advantage Standardization (SAS)**: Replacing the conventional stepwise advantage with a cumulative smoothed advantage substantially improves GRPO and brings it close to DAPO, indicating that aggregating reward statistics across training steps stabilizes optimization and improves final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: Replacing GRPO’s fixed clipping boundary with a probability-adaptive boundary yields pronounced gains; DAC alone surpasses GRPO, DAPO, and GSPO and is more stable than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: Combining all three mechanisms produces the strongest and most stable results, substantially outperforming GRPO, DAPO, and GSPO. The synergy among OTM, SAS, and DAC shows that each design choice contributes distinctly to data-efficient and stable reinforcement learning, improving both performance and robustness for large language models.

Overall, the ablation study confirms that each DCPO component contributes positively to aggregate performance and that their combination produces substantial cumulative gains. These results validate the proposed mechanisms as effective means of improving data efficiency and optimization stability in reinforcement learning for LLMs.

6 CONCLUSION

This work addresses two central limitations of existing policy-optimization methods for RLVR: restricted token-level exploration caused by fixed clipping in low-probability regions, and low sample efficiency induced by current-step reward standardization. To this end, we introduced **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement-learning pipeline that improves the reasoning capabilities of large language models through two complementary innovations: (i) a **dynamically adaptive token-level clipping** mechanism that adjusts bounds according to old-policy probabilities, thereby enabling more effective exploration of rare, low-probability tokens; and (ii) **Cumulative-Smooth Advantage Standardization**, which aggregates cumulative reward distributions across optimization steps and mitigates reward randomness from high-entropy sampling. Extensive experiments on four mathematical reasoning benchmarks and four model backbones demonstrate the effectiveness of DCPO. For example, DCPO-7B achieves an Avg@32 score of 38.8 on the challenging AIME24 benchmark, outperforming GRPO (32.1) and DAPO (31.6). In addition, DCPO increases the response utilization ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude. These results indicate that DCPO substantially improves data utilization and diversity exploration in RLVR. Future work will examine the extension of DCPO to additional domains, including code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In prior methods, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for token t in response j is computed by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards obtained from the G responses generated at the i -th step, as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When all rewards associated with the same prompt are identical, the resulting zero-advantage responses cannot contribute to model updates, producing response-level waste.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, which samples G responses for each query, assigns rewards R using a rule-based reward function, and estimates token-level advantages according to Equation (11). The model parameters are then updated with Equation (12), referred to as **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \| \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages first across all tokens within a response and then across all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty constrains the policy from deviating excessively from a reference policy (e.g., a pre-trained model), while the fixed clipping threshold ε limits token-probability-ratio changes to prevent overly large updates. Although widely adopted, this averaging strategy inherently dilutes per-response importance.

A.3.2 DAPO

Similarly, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**, motivated by the observation that longer GRPO sequences exert disproportionate influence on gradient updates. DAPO standardizes token counts over the entire batch, yielding the **Token Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. Prompts for which all responses receive identical rewards are discarded and regenerated to preserve batch size, which induces substantial data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can introduce high-variance noise into gradient estimates. Because this noise accumulates over long sequences and can be further amplified by clipping itself, GSPO replaces token-wise clipping with sequence-wise clipping to obtain a more stable and effective training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp\left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log\left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)}\right)\right)$$

GSPO discards more than 10% of responses with nonzero advantages, thereby producing substantial data waste and reducing training efficiency. Although GRPO avoids discarding responses solely because of sequence-level high variance, it still retains many responses with token-level high entropy, which can contribute to training instability.

A.4 REWARD CALCULATION

To supply a verifiable and interpretable supervision signal, we adopt a rule-based reward model that evaluates each generated response according to answer correctness and format compliance, as shown in Equation (15). Specifically, a reward of +1 is assigned when both the format and answer are correct, a reward of 0 when the format is correct but the answer is incorrect, and a reward of -1 when the format is invalid.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response group for a given problem at the i -th training step, and the reward is determined jointly by response format and answer correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Probability-ratio clipping is a crucial technique for stabilizing reinforcement-learning policy optimization and preventing large, destabilizing parameter updates. The mechanism is especially important when importance sampling is used to estimate expectations under a new policy from samples generated by an old policy.

The expectation of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ through importance weighting, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, it can suffer from substantial variance inflation, a standard difficulty in importance sampling. The true variance of $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} [f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x) \frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The gap between the importance-sampled variance and the true variance makes this inflation explicit, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To mitigate this discrepancy, many prior methods, including PPO Schulman et al. (2017) and RLVR algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025), impose a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

However, a fixed clipping bound intrinsically constrains the policy’s exploratory capacity, especially in regions where the old policy assigns low probability mass. In these cases, the absolute admissible variation for low-probability tokens is severely restricted, limiting meaningful policy updates. Although methods such as DAPO attempt to address this issue through asymmetric clipping, the more fundamental limitation of non-adaptive constraints remains.

In addition, recent studies such as Yue et al. (2025) suggest that the reasoning paths exploited by RLVR models are already largely contained within the base model’s sampling distribution. This claim is further supported by the small parameter shifts observed during off-policy batch updates, reflected in the limited fraction of clipped parameters. Given this stability, we propose a more practical alternative that constrains the probability ratio $r(x)$ through a dynamic-adaptive mechanism, incorporating probability directly into the restriction in Equation (21) and replacing the conventional fixed-clipping formulation of Equation (20).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality and expanding it yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to remain mathematically valid and interpretable, it must satisfy the fundamental constraints governing probability distributions and their ratios. As shown in Equation (23), $p(x)$ and $q(x)$ must be non-negative and at most 1; consequently, the probability ratio $r(x)$ and the clipping hyperparameter ϵ must also be non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To ensure that the clipping bound derived from Equation (22) is both valid and practically applicable, we state the necessary boundary conditions in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24) implies that the probability ratio $r(x)$ must satisfy the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we select the overlapping region centered at ratio value 1, both to minimize discrepancies between the two conditional bounds and to preserve consistency with GRPO's conventional fixed interval. This choice yields the following validity conditions:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we ensure that the square-root argument remains non-negative by comparing it with 0 and taking the maximum value, as in $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This produces the following **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we use distinct hyperparameters ϵ_{high} and ϵ_{low} so that the dynamic clipping interval aligns with GRPO at specific reference points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To align the dynamic-adaptive clipping bounds with the fixed symmetric bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as illustrated in Figure 4d, we derive the corresponding values of ϵ_{low} and ϵ_{high} by solving the equations in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases}
\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases} \quad (28)$$

In common prior practice(e.g., GRPO), the fixed clipping threshold ϵ is set to 0.2; under this setting, the corresponding dynamic-clipping hyperparameters are approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although $\epsilon_{low} \neq \epsilon_{high}$, this asymmetry serves a different purpose from DAPO’s asymmetric fixed clipping.

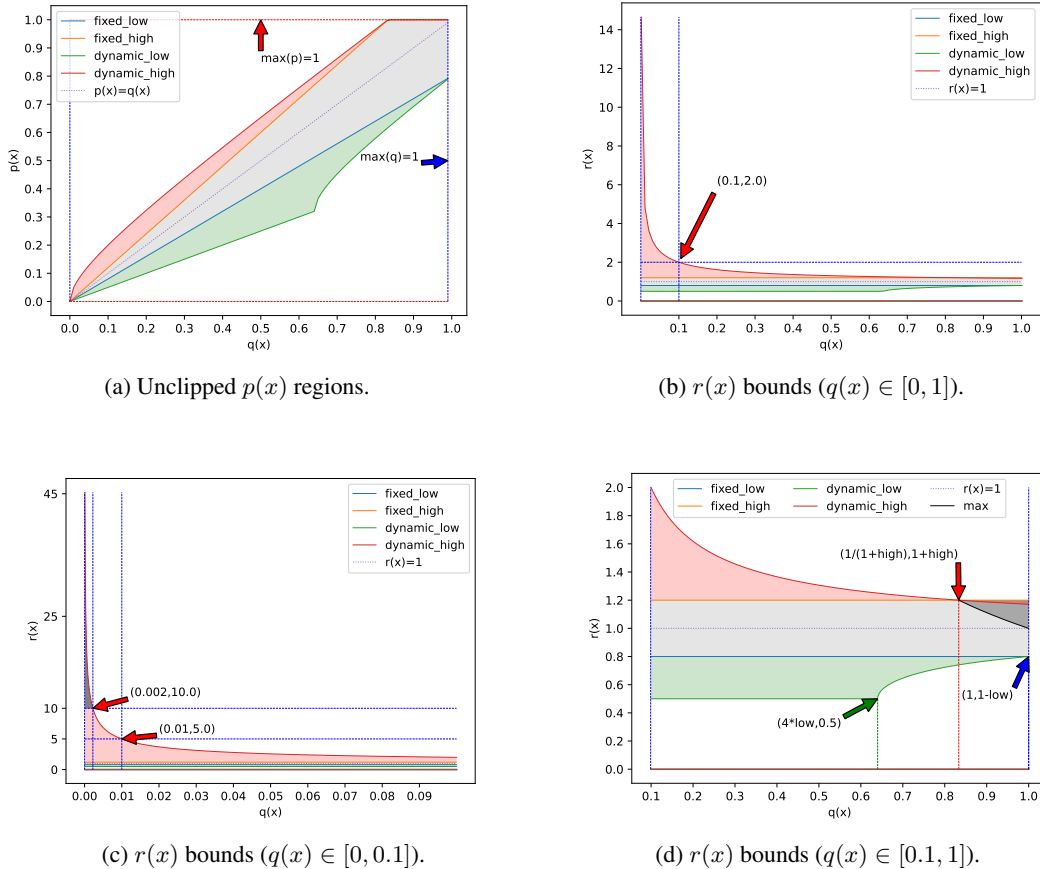


Figure 4: Comparison of clipping bounds, with lines contrasting fixed clipping ($\epsilon = 0.2$) and dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$).

Dynamic versus fixed clipping: Figure 4 contrasts the behavior of GRPO’s fixed clipping interval ($[1 - \epsilon, 1 + \epsilon]$) with the proposed probability-aware adaptive clipping. For a token whose old probability is $q(x)$, fixed clipping provides the constant relative width 2ϵ ; consequently, as $q(x)$ decreases, the admissible absolute exploration space shrinks linearly (Figure 4a). By contrast, the relative width induced by dynamic adaptation is inversely proportional to $q(x)$, which keeps the effective exploration space approximately constant across the probability range.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains independent of $q(x)$, so the model’s ability to explore low-probability tokens does not increase. The dynamic-adaptive bound, in contrast, expands rapidly (Figure 4b), creating the larger relative space for rare tokens that efficient RL requires.

Behavior for extremely low probabilities: Figure 4c displays the clipping interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound for $r(x)$ increases rapidly, with inverse square-root scaling, whereas the fixed-clipping counterpart remains constant. Inspired by the dual-clipping strategy (Ye et al., 2020), we impose a hard ceiling $r_{\max} = 10$ to prevent the resulting instabilities (gradient vanishing, explosion, etc.). The gray region in Figure 4c corresponds approximately to $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper limit is set to $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$, dynamic-adaptive clipping yields $dynamic_{low} = 0.5$, whereas fixed clipping retains the constant $fixed_{low} = 1 - \varepsilon = 0.8$. Once $q(x)$ exceeds this threshold, the dynamic-adaptive lower bound contracts smoothly toward the fixed value. Analogously, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$ and coincides with it for larger $q(x)$. Thus, the adaptive interval $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ widens the admissible region for rare tokens in a probability-dependent manner while preserving the tightness of the fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) includes both positive- and negative-advantage terms. Under the minimum-loss operation, the ratio $r(x)$ for a positive advantage is clipped to $[0, dynamic_{high}(q(x), \epsilon_{high})]$, whereas for a negative advantage the clipping interval is $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. Because excessively large values of $r(x)$ can destabilize training, we cap the ratio for both positive and negative advantages at $r_{\max} = 10$. This safeguard follows Ye et al. (2020) and prevents the adaptive limits from producing excessively high importance weights.

In short, for both positive and negative advantages, dynamic-adaptive clipping gives the model substantially more room than fixed clipping to explore diversity in low-probability rare-token regions.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration in which prompts generate responses and model parameters are updated. We define the following statistics for reward normalization.

For the response-reward means, we define the following group of functions in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

For the response-reward variances, we analogously define the group of functions in Equation (30):

$$\begin{cases} \sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\ \sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{old}^i)^2} \\ \sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce resource consumption when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent calculations based on μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i rather than directly recomputing statistics from the original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments in this paper use the following prompt template, which follows the Qwen-Math template (Yang et al., 2024b).

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

For GRPO, we set the symmetric clipping threshold to $\epsilon = 0.2$; for DAPO, we use lower/upper clipping thresholds (0.2, 0.28) together with an additional 512-token soft-punish cache and maximum probability ratio 10 for negative advantages; and for GSPO, we use lower/upper clipping thresholds $(3e - 4, 4e - 4)$. For DCPO, we set low/high clipping thresholds to (0.16, 0.2), preserving the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1 + \epsilon)$ and $(q(x), r(x)) = (1, 1 - \epsilon)$ as GRPO (Figure 4d), while setting the maximum probability ratio to 10 for both positive and negative advantages, unlike DAPO.

Training proceeds for 400 steps with a response-generation batch size of 512 and a parameter-update mini-batch size of 32. We use temperature 1 and top_p 1 to generate $G=16$ responses for each problem; the maximum prompt length is 1,024 and the maximum response length is 3,072, substantially smaller than the 20k maximum length used in the original DAPO.

We adapt the training codebase from Verl (Sheng et al., 2024) and run experiments on 32xH20 GPUs. Unless otherwise specified, hyperparameters follow DAPO. For loss computation, we retain the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulations from GRPO or DAPO, rather than the approximate weighted calculation used in Verl, to ensure compatibility across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports model performance under the averaged Avg@1 metric, computed by averaging greedy-decoding (Avg@1) results over MATH500, AMC23, AIME24, and AIME25 and thereby summarizing each model’s best deterministic performance. The figure shows that DCPO consistently outperforms GRPO and DAPO across all four models and also exceeds GSPO on the two models for which GSPO is evaluated. Moreover, DCPO’s overall trajectory is more stable than the baselines, avoiding both the pronounced performance degradation observed in GRPO and the large fluctuations observed in DAPO. This suggests that DCPO more effectively stimulates reasoning capability during training and creates greater scope for subsequent performance improvement.

Avg@32 Trend Figure 6 compares Avg@32 performance for DCPO, GRPO, DAPO, and, where available, GSPO across model sizes on AMC23, AIME24, and AIME25, highlighting DCPO’s robustness and stability under sampling. By contrast, GRPO exhibits substantial fluctuation and occasional performance collapse; with GRPO, Qwen2.5-14B even undergoes a marked performance decline after approximately 170 training steps. Across all evaluated models (from 1.5B to 14B), DCPO follows a consistently upward trajectory with minimal fluctuation, maintains the fastest early-stage growth rate, and continues improving thereafter. Relative to DAPO, DCPO also achieves stronger

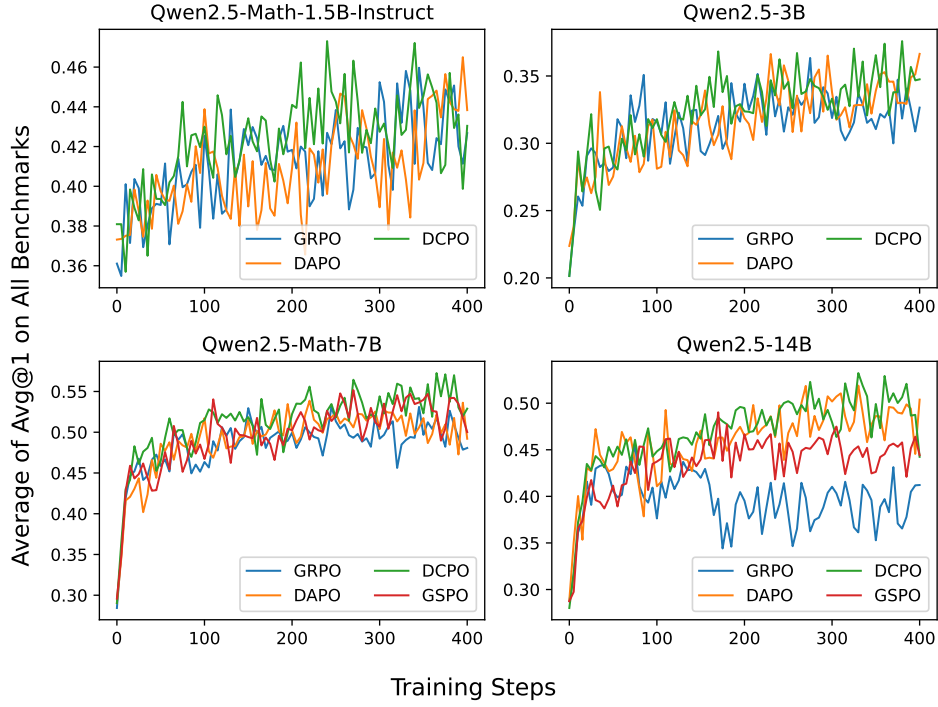


Figure 5: Avg@1 benchmark performance aggregated across tasks

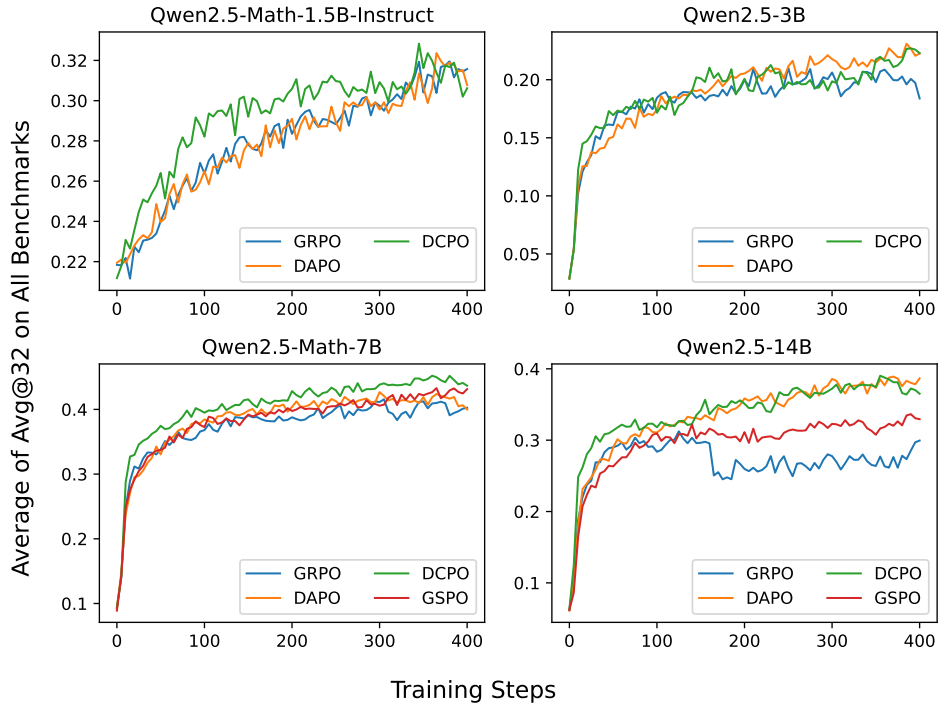


Figure 6: Avg@32 benchmark performance aggregated across tasks.

performance while requiring only half the training time and resources. These results demonstrate that dynamic-adaptive clipping and smooth advantage standardization jointly improve response-level exploration and label-level diversity in a robust manner.

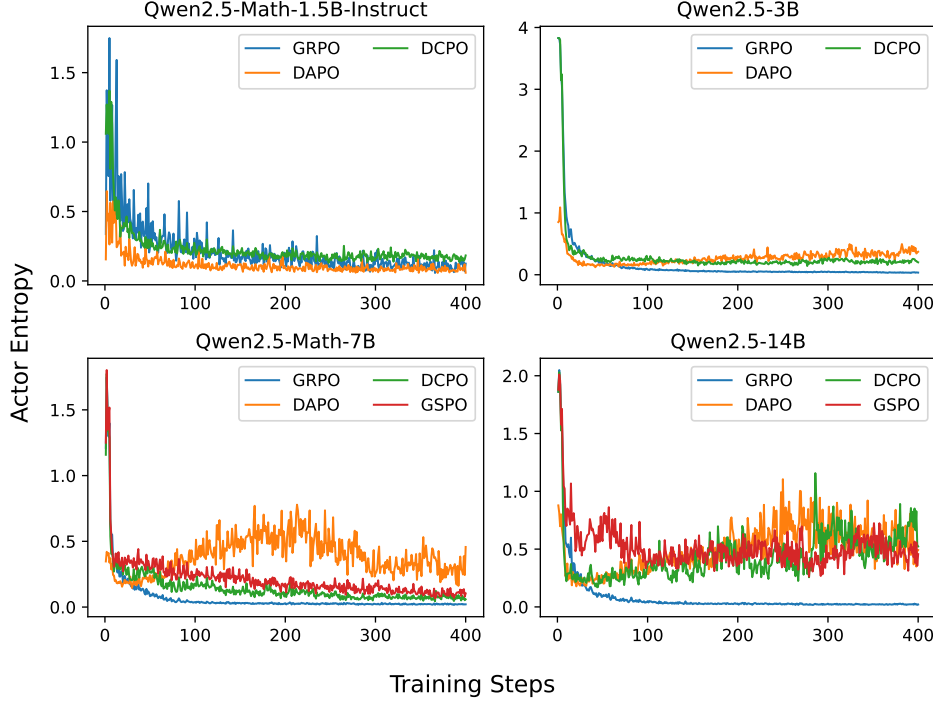


Figure 7: Model entropy trajectories during training.

Entropy Trend Figure 7 shows that, after an initial warm-up phase, GRPO undergoes a sharp entropy decline and rapidly stabilizes in a flat low-entropy regime. This entropy collapse indicates a rapid loss of policy randomness, which reduces sampled-response diversity and consequently limits subsequent policy optimization.

The behavior of DAPO and DCPO is model dependent. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow broadly similar declining trajectories. For the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), however, the patterns diverge: DAPO maintains the highest entropy, exhibiting a clear upward trend accompanied by large fluctuations. On Qwen2.5-Math-7B, GSPO’s entropy is similar to DCPO’s and its average metric performance is likewise comparable; on Qwen2.5-14B, however, GSPO displays a fluctuating downward performance trend and consequently performs poorly. DCPO instead settles at a moderate entropy level, with a substantially smoother curve bounded between GRPO’s low-entropy region and DAPO’s high-variance region.

These observations suggest a U-shaped relationship between entropy and training performance. Excessively low entropy deprives the learner of exploratory signal and induces premature convergence, whereas excessively high or unstable entropy compromises optimization stability. DCPO maintains entropy in a moderate, well-balanced range that supports both convergence and sufficient exploration, a benefit that becomes especially pronounced as model capacity increases. Accordingly, DCPO’s entropy-regulation behavior may help explain its superior inference performance and its robust policy learning across model scales.